Nikhil Bindal

Senior Full-Stack & AI Infrastructure Engineer

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SUMMARY

Senior engineer with 6+ years building and owning production distributed systems and applied-AI platforms across fintech, media, research, and early-stage startups. Took a multi-agent AI pipeline from 36s to 9.5s and a fintech underwriting flow from 8.7s to 890ms; designed event-driven backbones, low-latency RAG and voice systems, and the data layers behind them. Comfortable as a founding engineer: 0→1 ownership, system design under ambiguity, and calibrated, defensible engineering over buzzwords.

TECHNICAL SKILLS

Languages Python, TypeScript / JavaScript, Node.js, SQL

Backend & APIs FastAPI, Django, AsyncIO, Express, REST, WebSockets, gRPC, event-driven microservices

AI / ML Agentic orchestration (LangGraph, CrewAI, LlamaIndex), RAG & hybrid retrieval, embeddings, XGBoost, GPT-4o / Gemini, Vertex AI

Data & Messaging PostgreSQL (+ pgvector), Redis, Kafka, MongoDB, Neo4j, Qdrant, MySQL

Infrastructure AWS, GCP, Docker, Kubernetes, PgBouncer, Prometheus, Grafana, CI/CD

Architecture Distributed & event-driven systems, real-time streaming, idempotency & exactly-once design, caching, observability

Frontend React, Next.js, React Native, TypeScript, micro-frontends, D3.js

EXPERIENCE

AI Infrastructure Engineer (Contract) — Jan 2026 – Present

Neurologyca — AI coaching platform — *San Francisco, CA*

- **Architected a production multi-agent AI coaching platform** on GCP Cloud Run combining biometric/HCI signal processing with a persistent personal-memory system, owning system design end to end.
- **Cut end-to-end AI latency 73% (36s → 9.5s)** by re-architecting a 6-agent pipeline into a coordinator-orchestrated DAG with prompt consolidation, model tiering, and async parallel execution.
- **Designed "Mnemosyne," a context-scoped memory architecture** with semantic retrieval and a temporal-decay lifecycle, bucketed per user and domain so context can never cross boundaries.
- **Engineered Redis-backed concurrency control and session orchestration** that prevented state corruption across concurrent real-time voice sessions, plus a parallel voice sidecar (Gemini Live) running sentiment/query/analytics off the hot path.
- **Shipped Python and JavaScript SDKs (32 operations)** spanning orchestration, memory, analytics, sessions, and BYOD multi-tenant infrastructure, with CI/CD, audit logging, and webhook delivery.

Founding Engineer — Independent AI Venture (Stealth) — Jan 2024 – Dec 2025

Self-directed — building "Donna" & "RecoMe" — *Remote*

- **Built "Donna," an AI meeting-intelligence platform, end to end (0→1)** automating pre-meeting research, live in-meeting assistance, and post-meeting synthesis, and released it for real users.
- **Designed a 3-phase architecture orchestrating 15 specialized agents** behind a uniform agent contract that made parallel orchestration and fan-out trivial to extend.
- **Built hybrid contextual RAG over Qdrant** combining vector and keyword retrieval with reciprocal-rank fusion, improving retrieval relevance ~40% over a naive dense baseline (measured, not estimated).
- **Implemented a real-time voice pipeline (LiveKit WebRTC + Deepgram STT + Cartesia TTS)** targeting sub-200ms perceived latency, with a replica-independent WebSocket fan-out over Redis pub/sub.
- **Built "RecoMe," a personal interest-graph & agentic recommendation engine** — a capture → signal → graph → agent → surface pipeline turning cross-platform activity (15+ sources) into a typed Neo4j interest graph plus Qdrant per-user vectors, with recommendations streamed to the client over SSE.
- **Orchestrated 5 background agents behind a single Guardian gate** enforcing a hard $0.10/user/day LLM cost cap, throttling, and quiet hours, with a prompt-injection-resistant scorer (the LLM writes prose only, never the verdict); ran on BullMQ workers with idempotent Stripe metering and a dual consumer + multi-tenant IaaS surface on one backend (TypeScript, Hono, Prisma).

Full-Stack & AI Engineer — Feb 2023 – Dec 2023

Northeastern University — Minkara Computational Lab — *Boston, MA*

- **Built a semantic-search & visualization platform for biomedical research** (FastAPI, PostgreSQL + pgvector, React, D3.js) letting researchers search 10K+ papers and molecular datasets by meaning via embeddings and vector search.

- **Built AWS Batch distributed simulation pipelines** orchestrating thousands of parallel molecular computations and cutting compute cost ~40% via spot instances, right-sizing, and scale-to-zero with checkpoint/restart.
- **Owned backend for scientific data ingestion, metadata APIs, and ML-inference workflows** serving models as isolated services so heavy inference never blocked the API.
- **Designed accessibility tooling for visually impaired researchers** — tactile-graphics generation from molecular coordinates and screen-reader/sonification integrations, validated with blind users including the lab PI.

Software Engineer
Apr 2021 – Jul 2022

Times Internet — TOI+ subscription platform *Noida, India*

- **Built backend for TOI+ serving ~8.4M daily requests** — Node/Express subscription & paywall services with JWT auth and Redis-cached entitlements (sub-ms checks), keeping origin load low by offloading cacheable content to the Akamai edge.
- **Designed a Verdaccio-based micro-frontend system** packaging shared UI as versioned internal npm widgets, so 70+ city portals consumed updates by version bump instead of duplicated code — publish cadence decoupled from consume cadence.
- **Migrated 70+ legacy XML/XSLT portals to React micro-frontends** via a strangler-fig rollout (old + new in parallel, per-portal feature flags, stable API contracts), reaching ~92/100 Lighthouse with instant rollback.
- **Built the real-time Kafka event pipeline feeding the Signals personalization engine** (behavior events → user feature profiles, ~90s refresh), contributing to a measured ~9% CTR lift on recommendations.

Founding Software Engineer
Jan 2019 – Mar 2021

Progcap — collateral-free SMB lending fintech *New Delhi, India*

- **Owned the underwriting & transaction backbone** of a collateral-free lending platform — Node/Express microservices on an event-driven Kafka backbone, with PostgreSQL as the transactional system of record (+ append-only ledger) and MongoDB for high-write capture.
- **Cut decision latency 90% (8.7s → 890ms)** by parallelizing serial KYC/fraud/credit-score checks, trimming the hot path, and adding compound indexes plus Redis caching with connection pooling.
- **Integrated XGBoost credit scoring on alternative data as an isolated service** with timeouts and rule-based fallback, reducing false negatives ~19% (model + rules, measured on repayment cohorts).
- **Engineered effectively-once disbursement via idempotency** — guarded atomic state transitions, idempotency-keyed bank/NPCI calls, and partitioned Kafka consumers; load-tested to ~22K events/sec at the event tier.
- **Built a Python/Celery service for feature assembly, batch jobs, and reconciliation** against the append-only ledger as the source of truth.

Software Engineer
Aug 2017 – May 2018

LiveMedia / LiveChek — insurance telematics *New Delhi, India*

- **Built backend services and APIs for real-time telematics and behavioral analytics** processing driving-behavior and user-interaction streams for insurance workflows.
- **Contributed to early-stage product and architecture decisions** in a fast-moving startup environment.

SELECTED PROJECTS

- **Context-Bucket Knowledge Graph:** a document-ingestion system whose centerpiece is automatic context routing — on upload it decides append-to-existing vs. create-new “bucket” via embedding similarity with an LLM tie-breaker, so related documents enrich one graph and unrelated ones stay isolated. A LangGraph state machine (prepare → route → ontology → extract → finalize) drives a CrewAI crew and a per-bucket inferred ontology (no hardcoded types), with graceful degradation when no LLM key is present. Python, FastAPI, LangGraph, CrewAI, Neo4j, Postgres, React.
- **Manifold — geometric, low-LLM retrieval engine:** ingests arXiv papers (auto-fetch → PDF extraction → agentic extract/resolve/validate with edge-level provenance) into a knowledge graph, then minimizes LLM calls by doing entity resolution and relationship validation in embedding/rule space rather than per-step prompts — canonicalizing variants like “3DGS” → “3D Gaussian Splatting” and cutting ~60 raw mentions to ~42 deduped entities per paper. Retrieval uses HippoRAG-style Personalized PageRank; concepts live in hyperbolic (Poincaré) space for hierarchy-aware similarity; context is compressed via propositions + MMR before a single answer call. Exposes the graph to AI clients (e.g. Claude) as an MCP server for tool-based search/retrieval, and a PIPELINE_MODE=field|legacy switch A/B-benchmarks the LLM-call savings. TypeScript, Hono, Drizzle, PostgreSQL, React.

EDUCATION

- **M.Sc., Artificial Intelligence** — University of the Cumberlands (2024–2025)
- **M.Sc., Information Systems** — Northeastern University (2022–2024)
- **B.Tech., Computer Science & Engineering** — Kurukshetra University (2012–2016)